

Jamia Millia Islamia

A Central University

DSC-503: COMPUTER NETWORKS

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UNIT 1 : INTRODUCTION

- Introduction to Computer Networks,
- LAN,
- MAN,
- WAN,
- Uses of Computer Networks,
- LAN Technologies- Transmission Topologies, Access methods.
- Network Architecture, Protocol and standards,
- References Model: OSI-ISO and TCP/IP,
- Overview: Circuit switching, Message switching and Packet switching.

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Deterministic and Non-deterministic Signals

- Non-deterministic signals are random in nature hence they are called random signals.
- Random signals cannot be described by a mathematical equation.
- They are modelled in probabilistic terms.

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Elements of a communication system

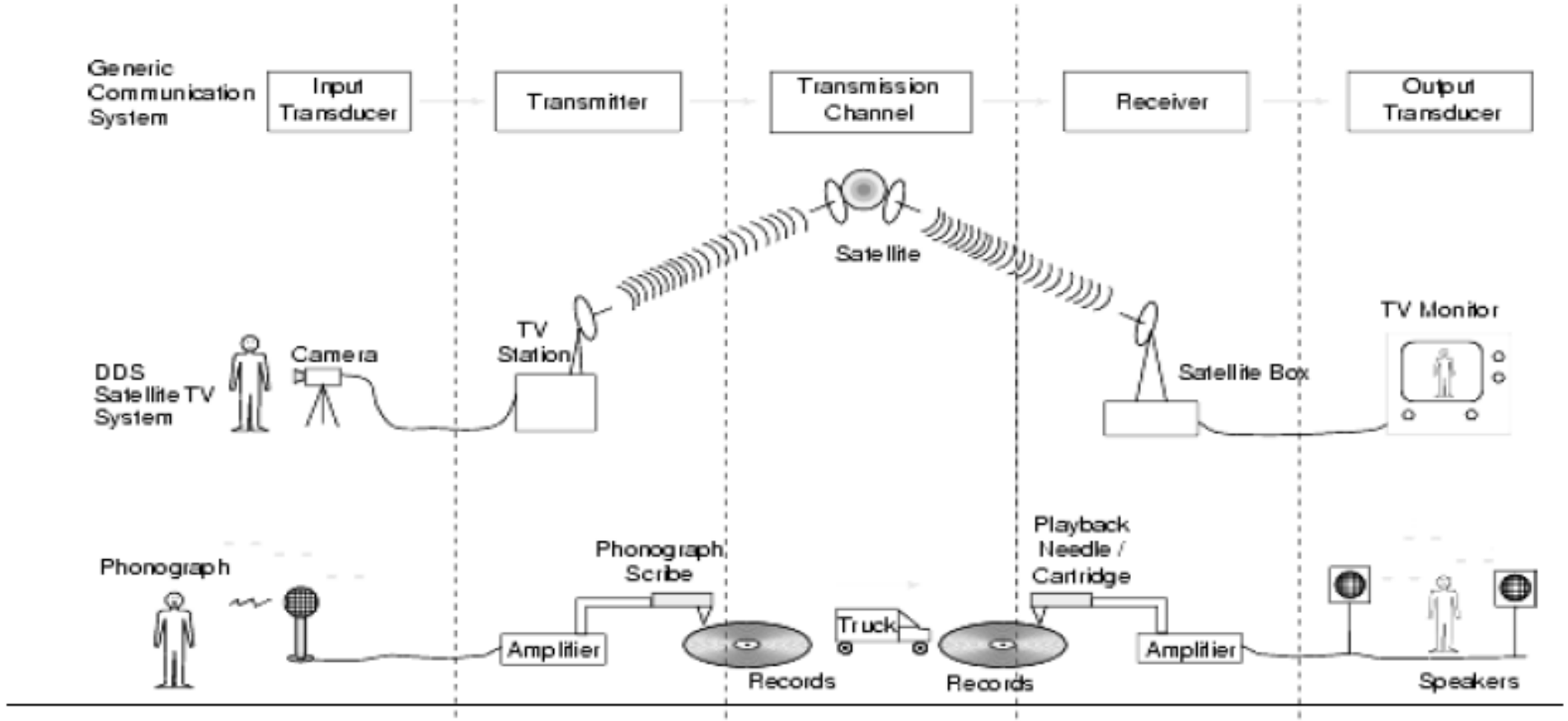


Image adapted From: *"Sharma, Sanjay. Data Communications "*

Elements of a communication system

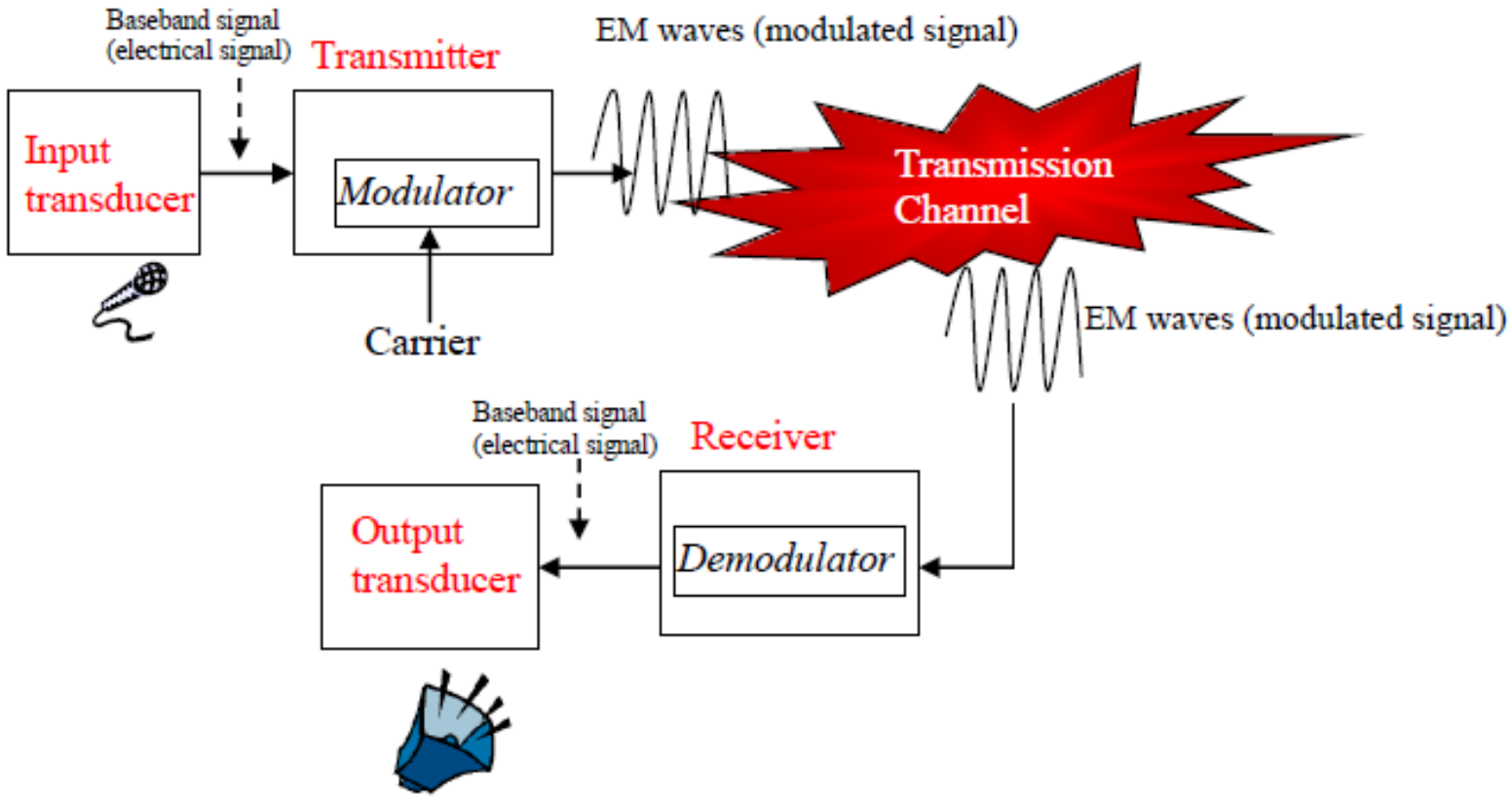


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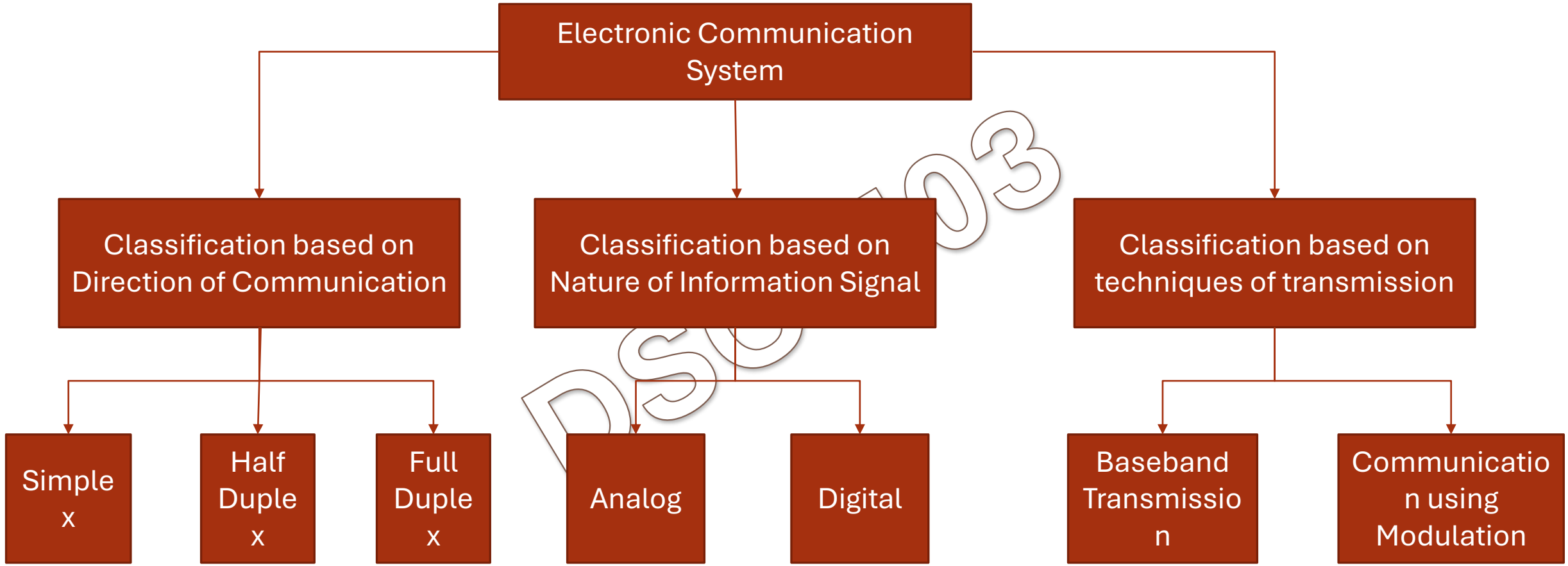
Classification of Electronic Communication System

- Electronic Communication can be broadly classified into three categories

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- Classification based on **Direction** of Communication.
- Classification based on **Nature** of Information Signal.
- Classification based on the **techniques** of transmission.

Classification of Electronic Communication System



Classification based on direction of Communication

- Electronic Communication
  - Unidirectional System: Simplex
  - Bidirectional System:
    - Half Duplex
    - Full Duplex

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Simplex System

- Simplex System: The information is communicated in one direction.

- E.g.
  - TV broadcasting,
  - Radio Broadcasting.

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Bidirectional System

- **Half Duplex Systems:** These systems are bidirectional.
- They can transmit and receive information but not simultaneously.

- **Full Duplex System:** These systems are bidirectional.
- They allow the communication to take place in both the directions simultaneously.

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Classification based on Nature of Information Signal

- **Analog Communication:**
- The technique in which one of the characteristics of the carrier is varied in proportion with the instantaneous value of modulating signal is called **Analog modulation System**.

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Classification based on Nature of Information Signal

- E.g.: Analog Modulation
  1. Amplitude Modulation(AM)
  2. Frequency Modulation (FM)
  3. Phase Modulation(PM)
  4. Pulse Amplitude Modulation (PAM)
  5. Pulse Width Modulation(PWM)
  6. Pulse Position Modulation(PPM)

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Analog Communication:

- **Advantage of Analog Communication**:
  - Transmitters and receivers are simple.
  - Low bandwidth
  - FDM(Frequency Division Multiplexing) can be used.
- **Applications:**
  - Radio Broadcasting
  - TV Broadcasting
  - Telephones

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Analog Communication:

- **Drawbacks:**
  - Noise effects the signal quality
  - Coding is not possible.
  - Not suitable for secret information transmission.

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Classification based on Nature of Information Signal

- **Digital Communication:**
- The modulation system or technique in which the transmitted signal is in the form of digital pulses of constant amplitude, constant frequency and phase is called as digital modulation system.

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Classification based on Nature of Information Signal

- E.g.:
- Pulse Code Modulation (PCM) and Delta Modulation (DM) are the examples of digital modulation.
  - In the PCM and DM, a train of digital pulses is transmitted by the transmitter.
  - All the pulses are of constant amplitude, width and position. The information is contained in the combination of the transmitted pulses.

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Digital Communication: Advantages

- **Advantages of digital Communication**:
- Due to digital nature of the transmitted signal, the interference of additive noise does not introduce many errors. Hence, digital communication has a better noise immunity.
- Due to the channel coding techniques used in digital communication, it is possible to detect and correct the errors introduced during the data transmission.

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Digital Communication: Advantages

- Repeaters can be used between transmitter and receiver to regenerate the digital signal. This improves the noise immunity further.

- Due to the digital nature of the signal, it is possible to use the advanced data processing techniques such as digital signal processing, image processing data compression etc.

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Digital Communication: Advantages

- TDM (Time Division Multiplexing) techniques can be used to transmit many voice channels over a single common transmission channel.
- Digital communication is useful in military applications where only a few permitted receivers can receive the transmitted signal.
- Digital communication is becoming simpler and cheaper as compared to the analog communication due to the invention of high speed computers and ICs.

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Digital Communication:

- **Drawbacks:**
  - The bit rates of digital systems are high. Therefore, they require a larger channel bandwidth as compared to analog systems.
  - Digital Modulation needs synchronization in case of synchronous modulation.

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Digital Communication:

- **Application of Digital Communication**
  - Long Distance communication between earth and space ships.
  - Satellite Communication
  - Military Communication Which needs coding
  - Telephone Systems
  - Data and Computer communications.

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Classification based on the techniques of transmission.

- **Baseband Transmission System:**
  - The baseband transmission systems, the baseband signals are directly transmitted.
  - The Baseband signal can be analog or digital.
  - The electrical equivalent of the original information signal is known as the baseband signal.

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Baseband Transmission System: Limitations:

- The baseband transmission cannot be used with certain mediums.
- It generally travels a small distance with some mediums.

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Communication Systems using Modulation

- **In the modulation Process:**
  i. The Modulating Signal
  ii. The carrier.

- **The Modulating Signal:** It is the baseband or information signal.
- **The carrier Signal:** It is a high frequency signal.

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Frequency

- Frequency is the rate of change w.r.t time.
- Frequency is defined as the number of cycles of a waveform per second.

$$f = \frac{1}{T}$$

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Frequency

- It is expressed in hertz (Hz).
- Change in a short span of time means high frequency.
- Change over a long span of time means low frequency.
- If a signal does not change at all, its frequency is zero.

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Two signals with the same amplitude and phase, but different frequencies

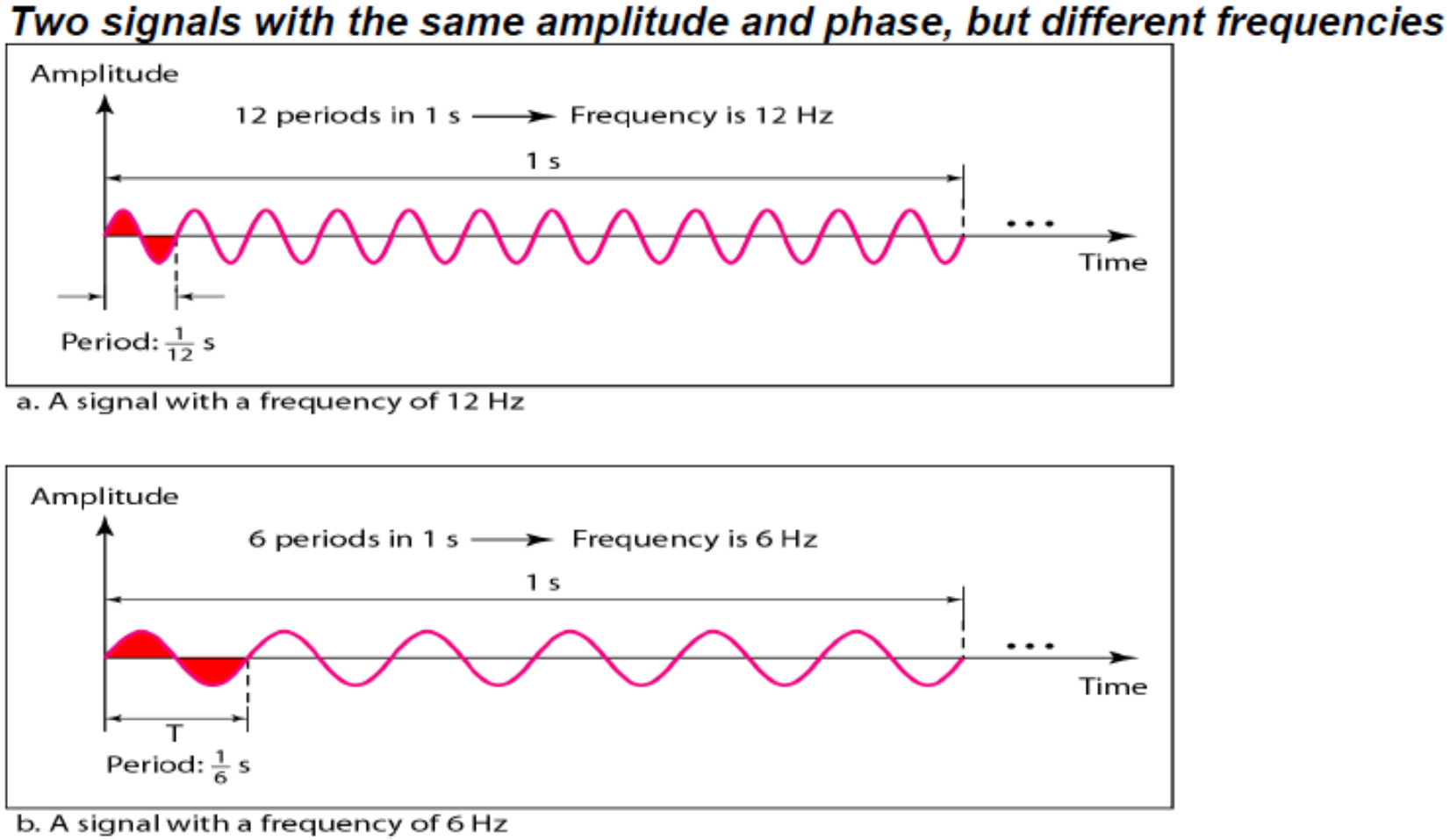


Image adapted From: *"Sharma, Sanjay. Data Communications "*

Frequency E.g.:

- ***Q1. The power we use at home has a frequency of 60 Hz. The period of this sine wave can be determined as follows:***

- Solution: f=1/T➔ T=1/f

Here f= 60 Hz.

T= 1/60= 0.0166s = 16.6 ms.

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Modulation

- In the modulation process, some parameters of the carrier wave is varied in accordance with the modulating signal.
- This modulated signal is then transmitted by the transmitter.

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Need of Modulation

I. TO REDUCE THE ANTENNA HEIGHT.

II. TO MULTIPLEX THE MORE NUMBER OF SIGNALS.

III. TO REDUCE THE NOISE & DISTORTIONS.

IV. TO NARROW BANDING THE SIGNAL.

V. TO REDUCE EQUIPMENT COMPLEXITY.

VI. TRANSMIT THE INFORMATION TO MORE DISTANCE WITHOUT INTERFERENCE.

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1. To Reduce The Antenna Height.

- When free space is used as a communication media, message are transmitted and received with the help of antennas.

- The message signal is radiated by an antenna at the transmitter.

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1. To Reduce The Antenna Height.

- The antenna radiates effectively when its height is of the order of the wavelength of the signal being transmitted.

- In broadcast systems, the maximum audio frequency transmitted from a radio station is of the order of 5KHz.

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Antenna Height

- If this signal were to be transmitted without modulation, the height of antenna needed for an effective radiation would be half of the wavelength given as :

$$\frac{\lambda}{2}$$

and

$$\lambda = \frac{c}{f}$$

Where c = speed of light and

f= frequency of signal

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Antenna Height for Baseband signal

- Minimum Antenna Height for a baseband signal of 5KHz can be calculated as:

$$\text{Minimum Antenna Height} = \frac{\lambda}{2} = \frac{c}{2f}$$

$$= \frac{3\ X\ 108}{2\ X\ 5\ X\ 103} = 30000\text{m} = 30\text{Km}$$

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Antenna Height for Baseband signal

- It would be impracticable to construct and install such an antenna.

- However, the height of the antenna can be reduced by analog modulation technique and yet achieve effective radiation.

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Antenna Height using Modulated Signal

- As modulation provides frequency shifting or frequency translation, audio frequency signals at radio stations(3Hz- 5KHz) are translated to higher frequency spectrum(radio frequency ranges).
- The higher radio frequencies with smaller wavelength act as carrier of the audio frequencies (modulating signal).
- Therefore, the height of the antenna required is much reduced and becomes practical.

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Antenna Height using Modulated Signal

- For example if an audio frequency is translated to a radio frequency carrier of frequency 1MHz, the antenna height required will be

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$$\text{Minimum Antenna Height} = \frac{\lambda}{2} = \frac{c}{2f}$$

$$= \frac{3\ X\ 108}{2\ X\ 1\ X\ 106} = 150\text{m}$$

- This antenna height can be practically achieved.

2. To Multiplex the more number of signals

- Simultaneous transmission of multiple messages over a channel is known as multiplexing.
- If transmitted without modulation, the different message signals over a single channel will interfere with one another.
- Different message signals can be transmitted over a single channel without interference using multiplexing techniques.

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2. To Multiplex the more number of signals

- Multiplexing helps in transmitting a number of messages simultaneously over a single channel and therefore the number of channels needed will be less.

- This reduces the cost of installation and maintenance of more channels.

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3. To Narrow Band the Signal

- There is another problem caused by direct radiation of baseband signal.
- Let us assume that baseband signal in a broadcast system is radiated directly with the frequency range extending from 50 Hz to 10KHz, the ratio of highest to lowest wavelength is 200.
- If an antenna is designed for 50 Hz, it will be too long for 10KHz and vice versa.

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To Narrow Band the Signal

- However, suppose the audio signal is translated to radio range frequency say 1 MHz then ratio of lowest to highest frequency will be

$$\frac{10^6 + 50}{10^6 + 10^4} \simeq \frac{1}{1.01}$$

which is approximately unity; and the same antenna will be suitable for the entire band

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To Narrow Band the Signal

- Thus frequency translation converts a wideband signal to a narrowband.
- Here bandwidth is referred to as the ratio of the edge frequencies of the band.
- This is called narrow banding.

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4. Avoid Mixing of Signals

- If the baseband sound signals are transmitted without using the modulation by more than one transmitter, then all the signals will be in the same frequency range i.e. 0 to 20 KHz.
- Therefore, all the signals get mixed together and a receiver cannot separated them from each other.
- If each baseband sound signal is used to modulate a different carrier then they will occupy different slots in the frequency domain/different channels.

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Fig: Modulation avoids mixing of signals

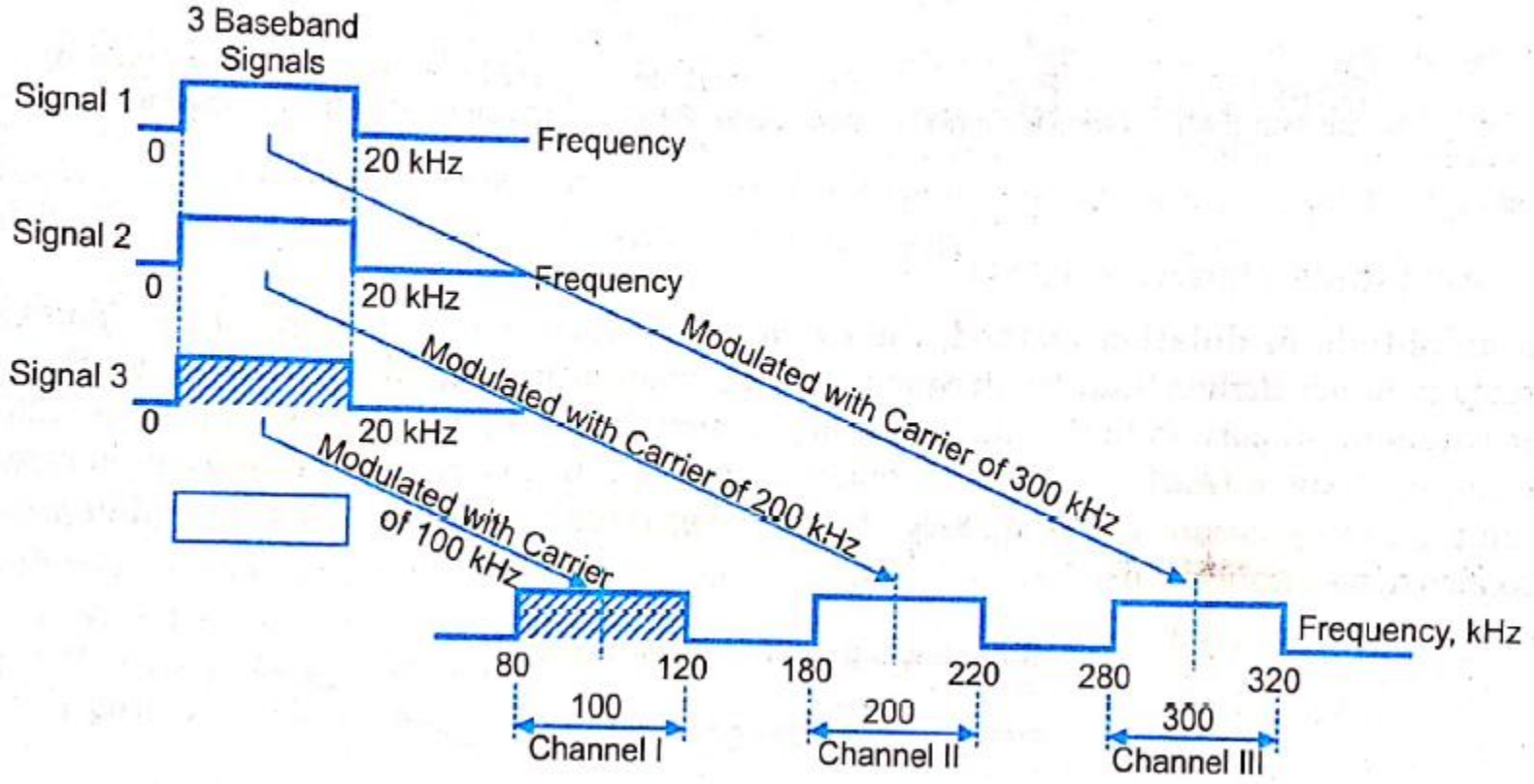


Image adapted From: *"Sharma, Sanjay. Data Communications "*

5. Improves quality of reception

- With frequency modulation (FM), and the digital communication techniques like PCM, the effect of noise is reduced to a great extent.
- This improves quality of reception.

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Electromagnetic Spectrum (EM)

- The information signal should be first converted into an electromagnetic signal before transmission.
- Wireless communication takes place using the EM waves.
- EM consists of
  - Electric fields
  - Magnetic fields
- EM waves can travel through long distance through space.
- EM waves are also called as Radio frequency (RF) Waves.
- The entire range of frequencies of EM waves is called as Electromagnetic spectrum

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Electromagnetic spectrum

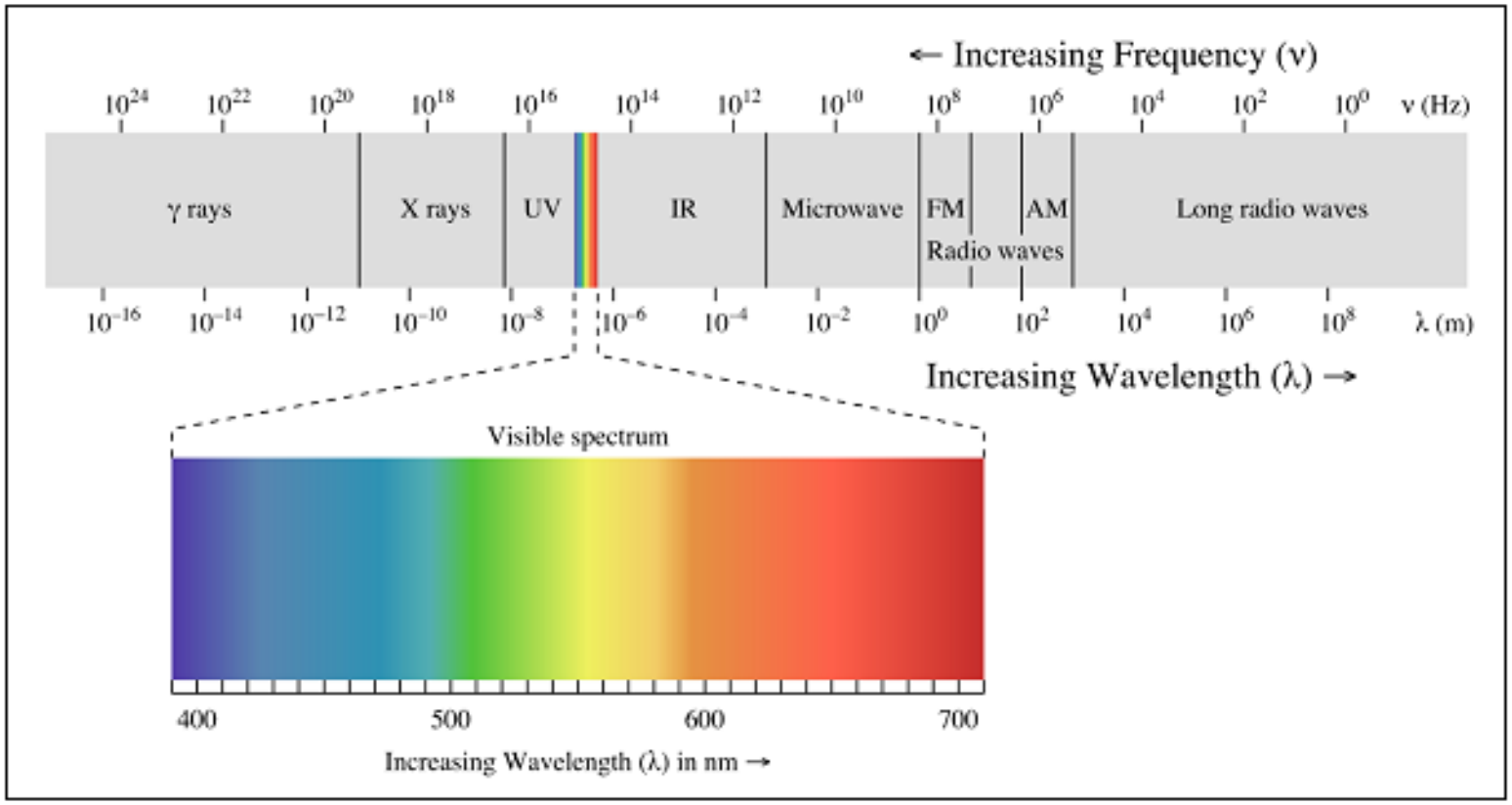


Image adapted From: *"Sharma, Sanjay. Data Communications "*

Complete EM Spectrum

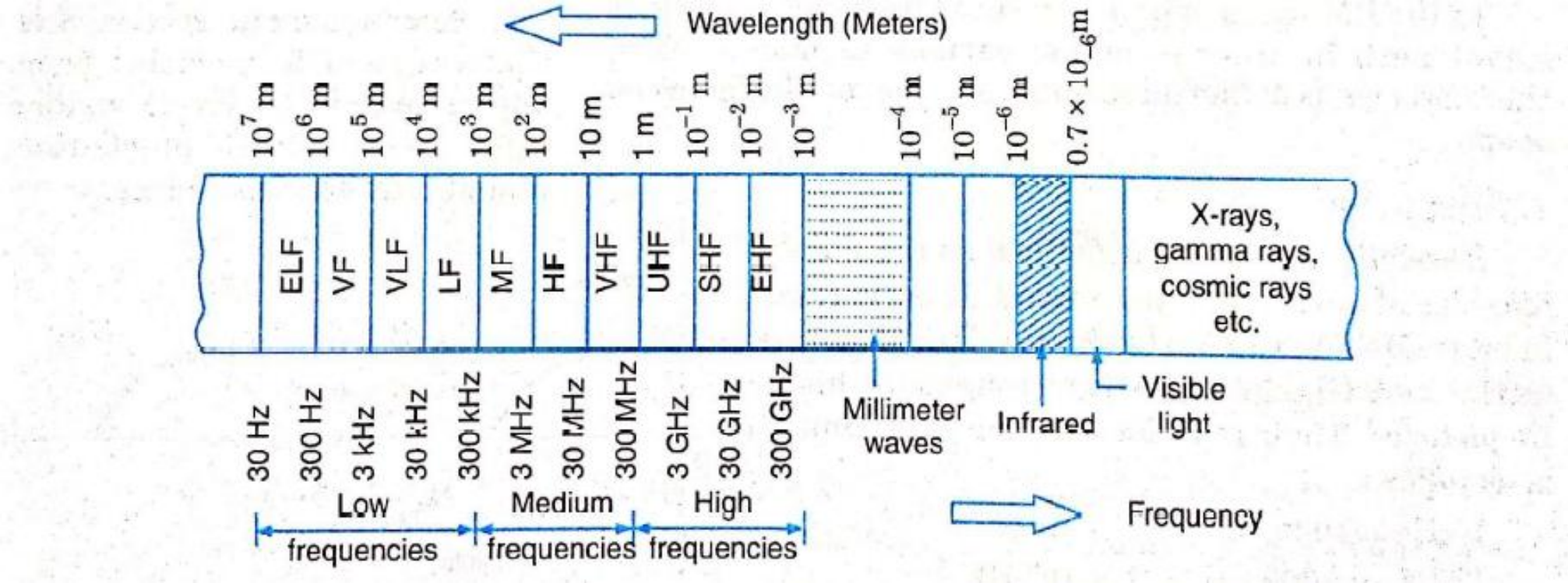


Image adapted From: *"Sharma, Sanjay. Data Communications "*

Frequency and Wavelength

- Frequency (f) is the rate of repetition of a regular event.
- Wavelength (λ) is defined as the distance between two points of similar cycles of a periodic wave.

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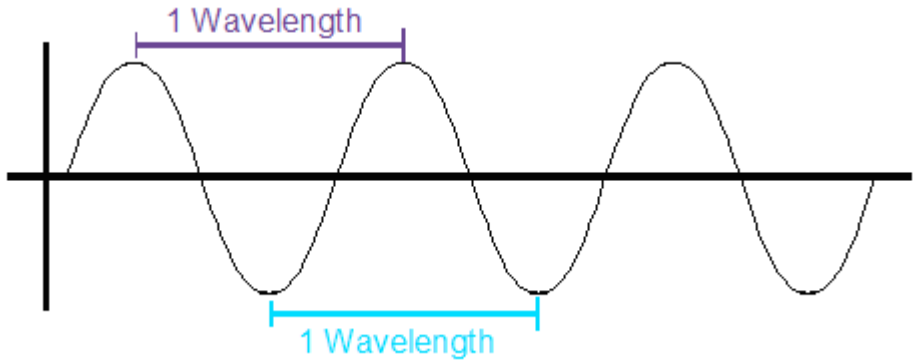


Image adapted From: *"Sharma, Sanjay. Data Communications "*

Wavelength

- Wavelength may also be defined as the distance travelled by an electromagnetic wave during the time of one cycle.
- **Wavelength** is defined as the distance from a particular height on the wave to the next spot on the wave where it is at the same height and going in the same direction.
  - Usually it is measured in m, just like any length.

$$\lambda = \frac{c}{f}$$

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Segment of EM Spectrum

S.No.	Name	Frequency	Wavelength
1.	Extremely low frequency (ELF)	30-300 Hz	10^7 to 10^6 m
2.	Voice frequecies (VF)	300-3000 Hz	10^6 to 10^5 m
3.	Very low frequencies (VLF)	3-30 kHz	10^5 to 10^4 m
4.	Low frequencies (LF)	30-300 kHz	10^4 to 10^3 m
5.	Medium frequencies (MF)	300 kHz-3 MHz	10^3 to 10^2 m
6.	High frequencies (HF)	3-30 MHz	10^2 to 10 m
7.	Very high frequencies (VHF)	30-300 MHz	10 to 1 m
8.	Ultra high frequencies (UHF)	300 MHz-3GHz	1 to 10^{-1} m
9.	Super high frequencies (SHF)	3-30 GHz	10^{-1} to 10^{-2} m
10.	Extremely high frequencies (EHF)	30-300 GHz	10^{-2} to 10^{-3} m
11.	Infrared	—	0.7 to 10 μm
12.	Visible light	—	0.4 μm to 0..8 μm

Image adapted From: *"Sharma, Sanjay. Data Communications "*

Radio Spectrum:

- All FM radio stations transmit in a **band** of frequencies between 88 megahertz and 108 megahertz.
- This band of the radio spectrum is used for no other purpose but FM radio broadcasts.
- AM Radio is confined to a band from 535 kilohertz to 1,700 kilohertz

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The Radio Frequency (RF)

S.No.	Frequency band	Wavelength	Applications
1.	30 Hz – 300 Hz. Extremely low frequencies ELF.	10^4 km to 10^3 km	Power transmission
2.	300 Hz – 3 kHz. Voice frequencies (VF)	10^3 km to 100 km	Audio applications
3.	3 kHz – 30 kHz Very low frequencies (VLF)	100 km to 10 km	Submarine communications. Navy, Military communications
4.	30 kHz – 300 kHz. Low frequencies (LF)	10 km to 1 km Long waves.	Aeronautical and marine. navigation, these frequencies act as sub carriers.

Image adapted From: *"Sharma, Sanjay. Data Communications "*

The Radio Frequency (RF)

5.	300 kHz – 30 MHz Medium frequencies (MF)	1 km to 100 m. Medium waves.	AM radio broadcast, Marine and aeronautical communications.
6.	3 MHz –30 MHz High frequencies (HF)	100 m to 10 m. Short waves	Shortwave transmission, Amateur and CB communication.
7.	30 MHz – 300 MHz Very high frequencies (VHF)	10 m to 1 m	T.V. broadcasting FM broadcasting
8.	300 MHz – 3 GHz Ultra high frequencies (UHF)	1 m to 10 cm. Microwaves.	UHF T.V. channels, Cellular phones, Millitary applications.
9.	3 GHz – 30 GHz (SHF)	10^{-1} m to 10^{-2} m	Satellite communications and Radar
10.	30 – 300 GHz (EHF)	10^{-2} m to 10^{-3} m	Satellites and specialized radars.

Image adapted From: *"Sharma, Sanjay. Data Communications "*

Infrared

- The EM signals having frequencies above 300 GHz are not referred as radio waves.
- The signal occupy the range between 0.1mm ad 700 nanometers are called infrared signals.
- Application:
  - Astronomy
  - Weapon Guidance
  - T.V. Remote systems

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Refer Class Notes and Recommended Books for further Reading

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Reference Books

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Thank You

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